

1. Project Development Duration

The FitBuddy project is planned as a 5-day development cycle covering requirement finalization, backend and database development, frontend development, integration, testing, documentation, and final demonstration preparation.

2. Day-wise Development Plan

3. Project Modules

4. Team Responsibility

5. Integration Plan

Integrate the frontend user-input flow with the FastAPI backend.

Connect the backend with the AI-based workout-plan generation functionality.

Connect application operations with the database for storing required information.

Integrate the feedback flow with updated-plan generation.

Verify the complete application workflow after module integration.

6. Testing and Final Submission Plan

Perform input-validation and functional testing.

Test FastAPI endpoints using Swagger UI.

Verify AI-based workout-plan generation and output display.

Verify feedback submission and updated-plan generation.

Check database storage and retrieval.

Capture relevant application screenshots as evidence.

Prepare the final project documentation and demonstration materials.

7. Expected Project Workflow

Project Planning → Development → Module Integration → Functional Testing → Documentation → Final Demonstration

8. Phase 4 Conclusion

The Phase 4 planning defines the project development schedule, module allocation, integration activities, testing activities and final submission preparation for FitBuddy. The plan provides a structured sequence for completing and demonstrating the project.

Day	Planned Activities	Team Responsibility
Day 1	Finalize project idea, requirements, modules, database design and UI design	Entire Team
Day 2	Develop backend, database and core application modules	Team

Day	Planned Activities	Team Responsibility
Day 3	Develop frontend pages and integrate workout-plan generation features	Team
Day 4	Integration, debugging and functional testing	Entire Team
Day 5	Final testing, documentation, screenshots and project demonstration preparation	Entire Team

Module	Planned Responsibility
User Input Module	Collects the user information required for fitness-plan generation.
Workout Generation Module	Generates the personalized 7-day workout plan using the application's AI-based generation feature.
Nutrition and Recovery Module	Provides general nutrition and recovery suggestions along with the workout plan.
Feedback Module	Collects user feedback about the generated plan.
Plan Update Module	Generates an updated plan based on the original plan and user feedback.
Database Module	Stores user details, plans, feedback and updated plans.
API Module	Provides FastAPI routes and supports API testing through Swagger UI.

Team Member	Assigned Topics / Responsibility
Annalakshmi	Pre-requisites, Development Environment, app.py, Conclusion
Maheswari	Workflow, FastAPI backend, Jinja2
Sakthi	Gemini model, deployment, testing
Sivajothi	Architecture, core functionality, UI